Zine Machine main board v3 (bancouver40_pb2) — generated by tools/zm_v3/gen_sch.py from design.py; edit design.py and regenerate.
Blocks: PocketBeagle 2 Industrial (face-down on its 3.55 mm sockets) + 4.58in RGB565 bar panel; 43 Choc v1 keys read by the
Cortex-M4F through six 74HC165 (CLK P1.26, SH/LD P1.28, QH P2.05); FE1.1s hub on USB1 -> 2 rear USB-A; IP5306 1S charger/boost
with a DPDT slide switch (P-FET on +5V_SYS + one KEY press); ES8388 codec (I2S master) -> headset TRRS + line-out TRS.
Software: header pins with two balls (P1.04/06/12/26/28, P2.01/03/05/07/09/11) — mux only the ball named here, keep the twin an input.

PocketBeagle 2 + display

pb2_display.kicad_sch

Keys and scan chain

keys.kicad_sch

USB hub and power

usb_power.kicad_sch

Audio

audio.kicad_sch

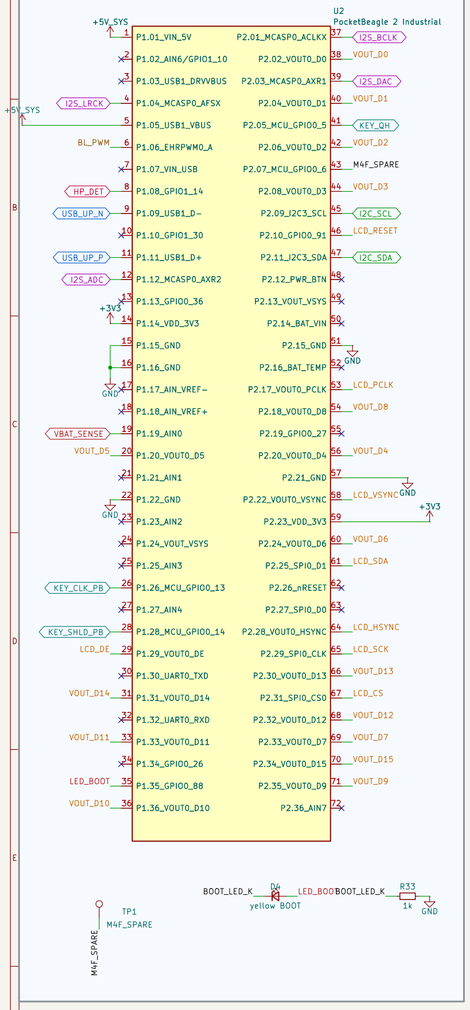
Generated by tools/zm_v3/gen_sch.py from design.py
bancouver40 + PocketBeagle 2 Industrial + 4.58in bar display

Sheet: /
File: bancouver40_pb2.kicad_sch

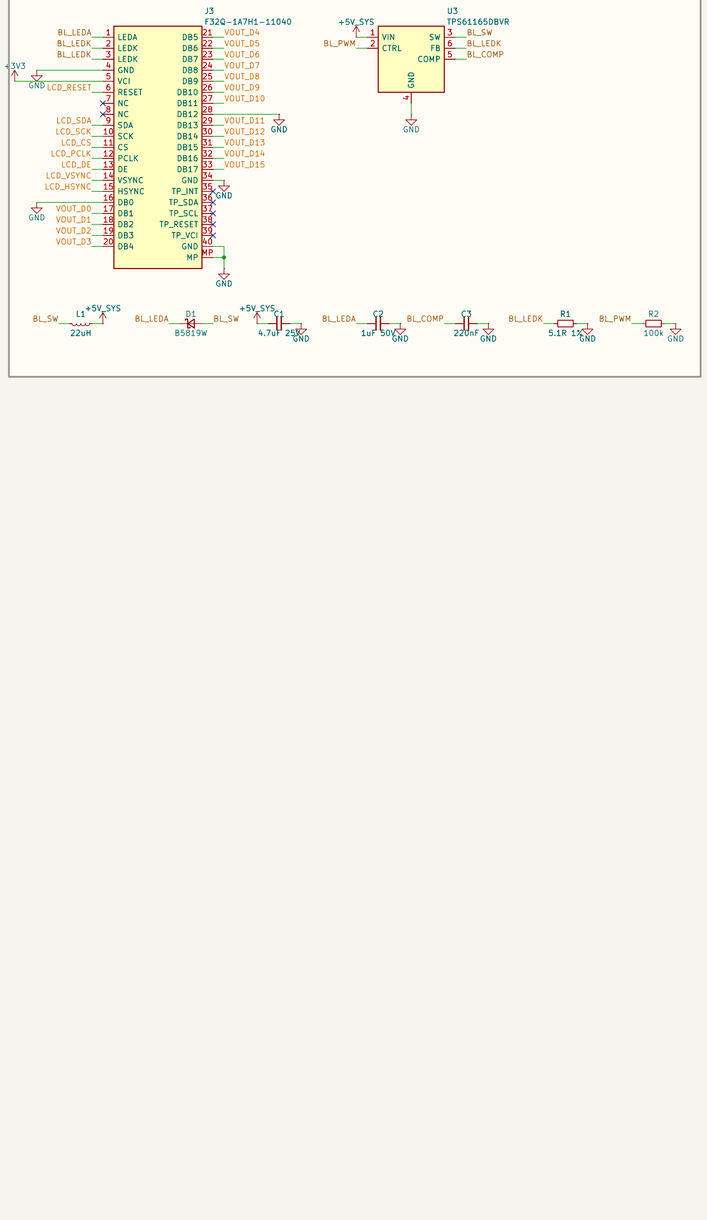
Title: Zine Machine main board v3: overview

Size: A3	Date: 2026-11-06	**Rev: 3**
KiCad E.D.A. backplane-kicad-v10.99.0-bp.6		Id: 1/5

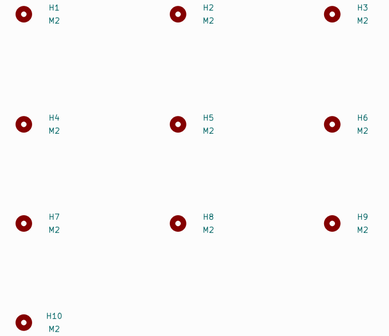
PocketBeagle 2 Industrial (face-down, USB-C to the rear)



4.58" bar panel FPC + TPS61165 backlight (PWM on P1.06)



Mounting holes



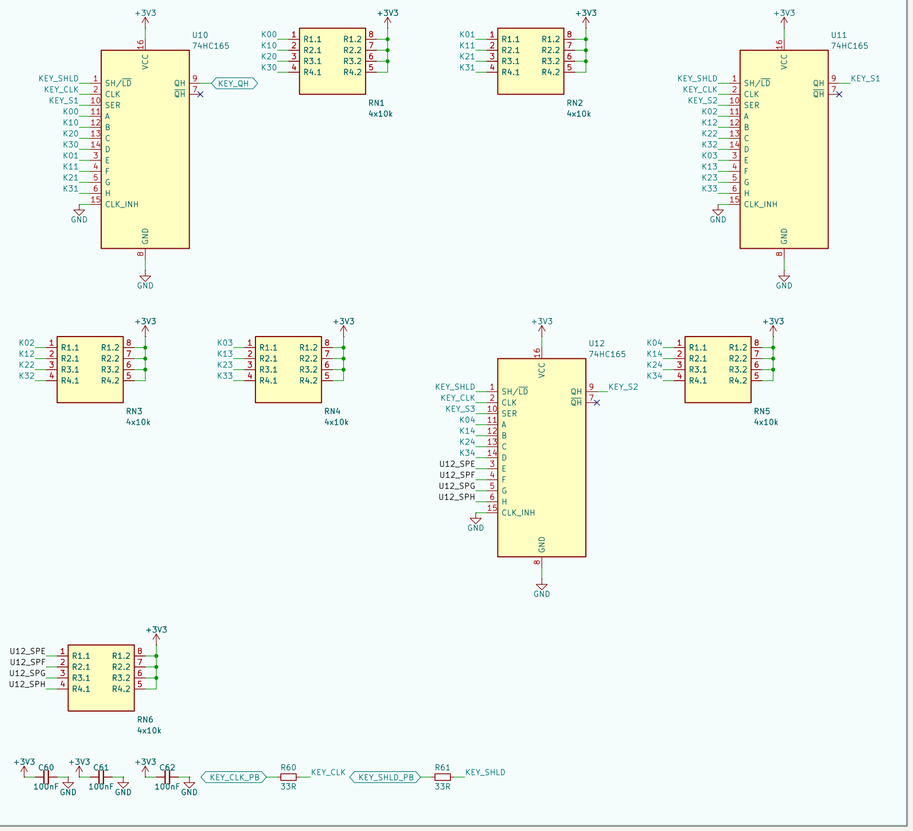
Generated by tools/zm_v3/gen_sch.py from design.py
bancouver40 + PocketBeagle 2 Industrial + 4.58in bar display

Sheet: /PocketBeagle 2 + display/
File: pb2_display.kicad_sch

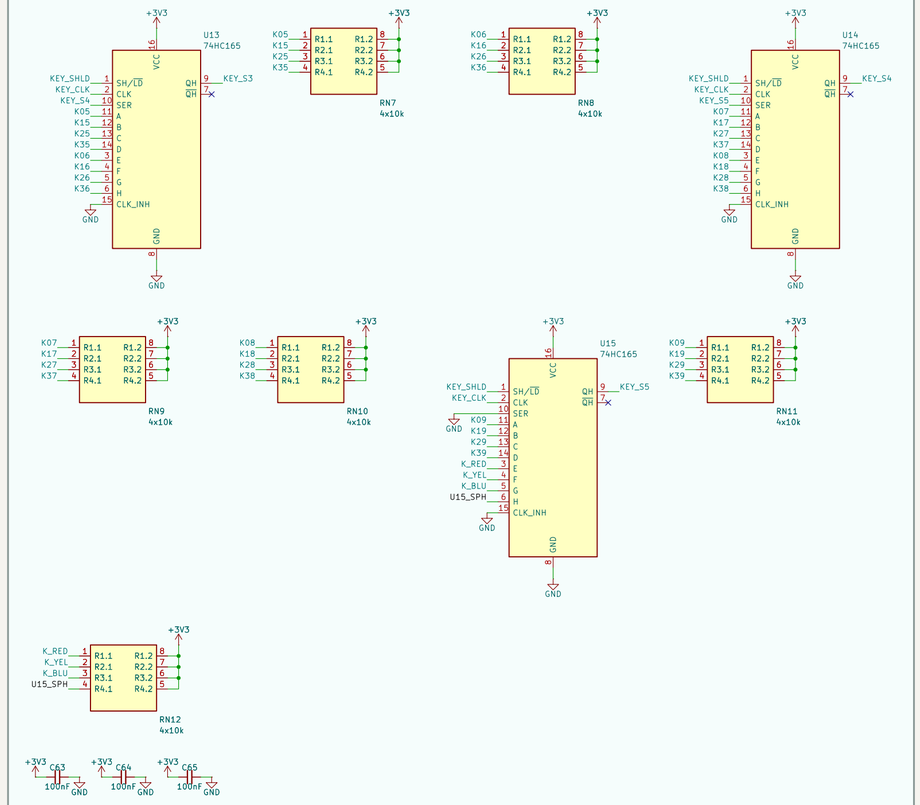
Title: Zine Machine main board v3: PocketBeagle 2 + display

Size: A2	Date: 2026-11-06	**Rev: 3**
KiCad E.D.A. backplane-kicad-v10.99.0-bp.6		Id: 2/5

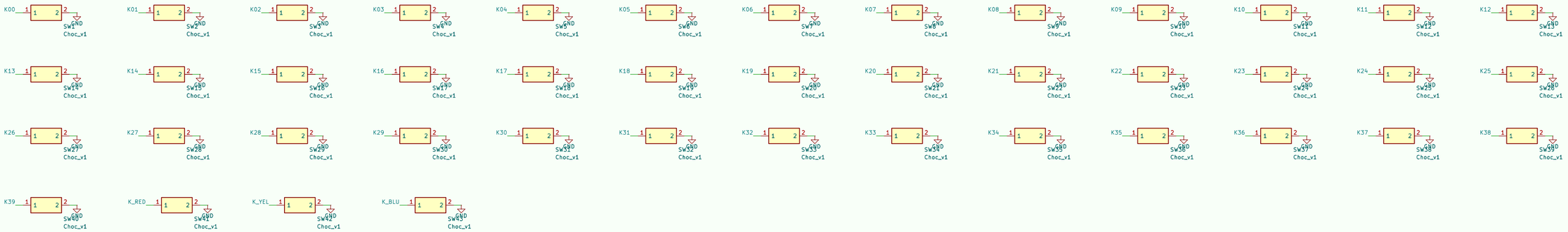
Key scan chain, left: U10–U12 (under the PocketBeagle)



Key scan chain, right: U13–U15 (+ mode keys)



Keys: 40 Choc v1 + 3 mode keys, wired direct to the 165 chain (key to GND)



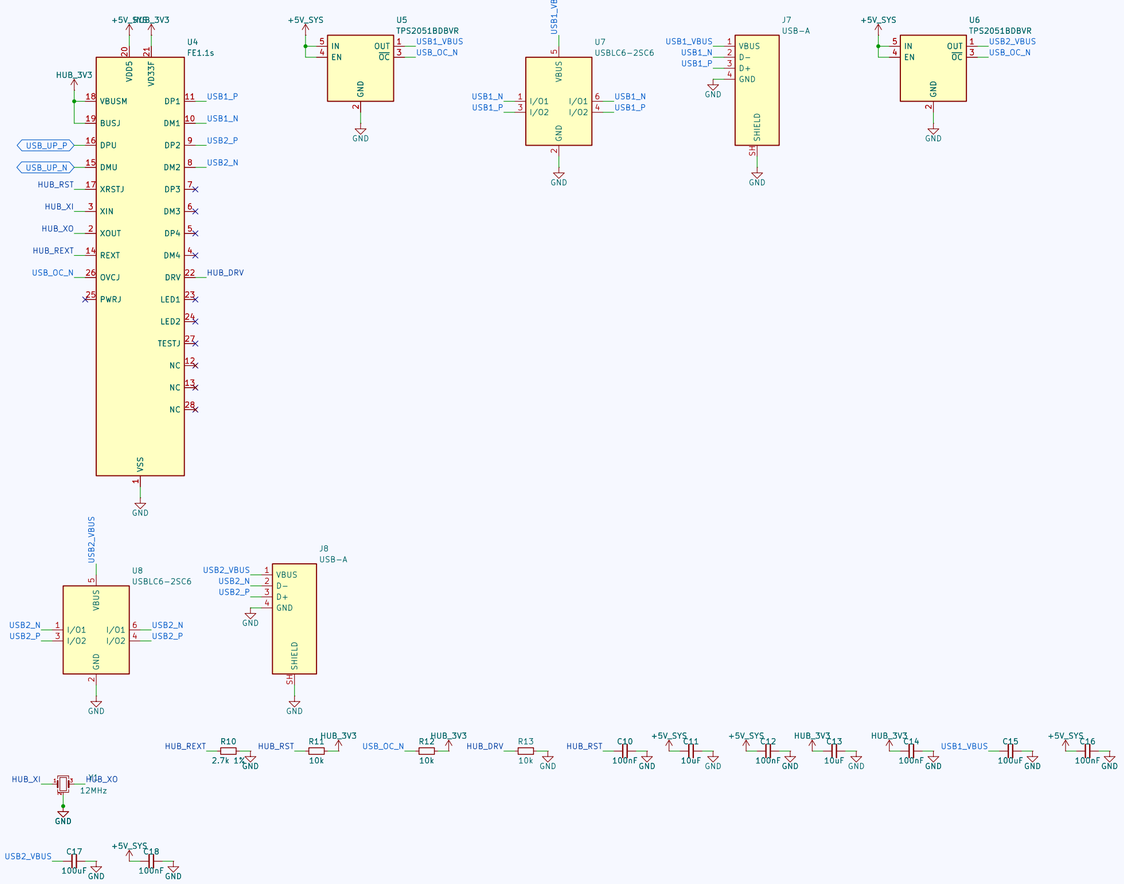
Generated by tools/zm_v3/gen_sch.py from design.py
bancouver40 + PocketBeagle 2 Industrial + 4.58in bar display

Sheet: /Keys and scan chain/
File: keys.kicad_sch

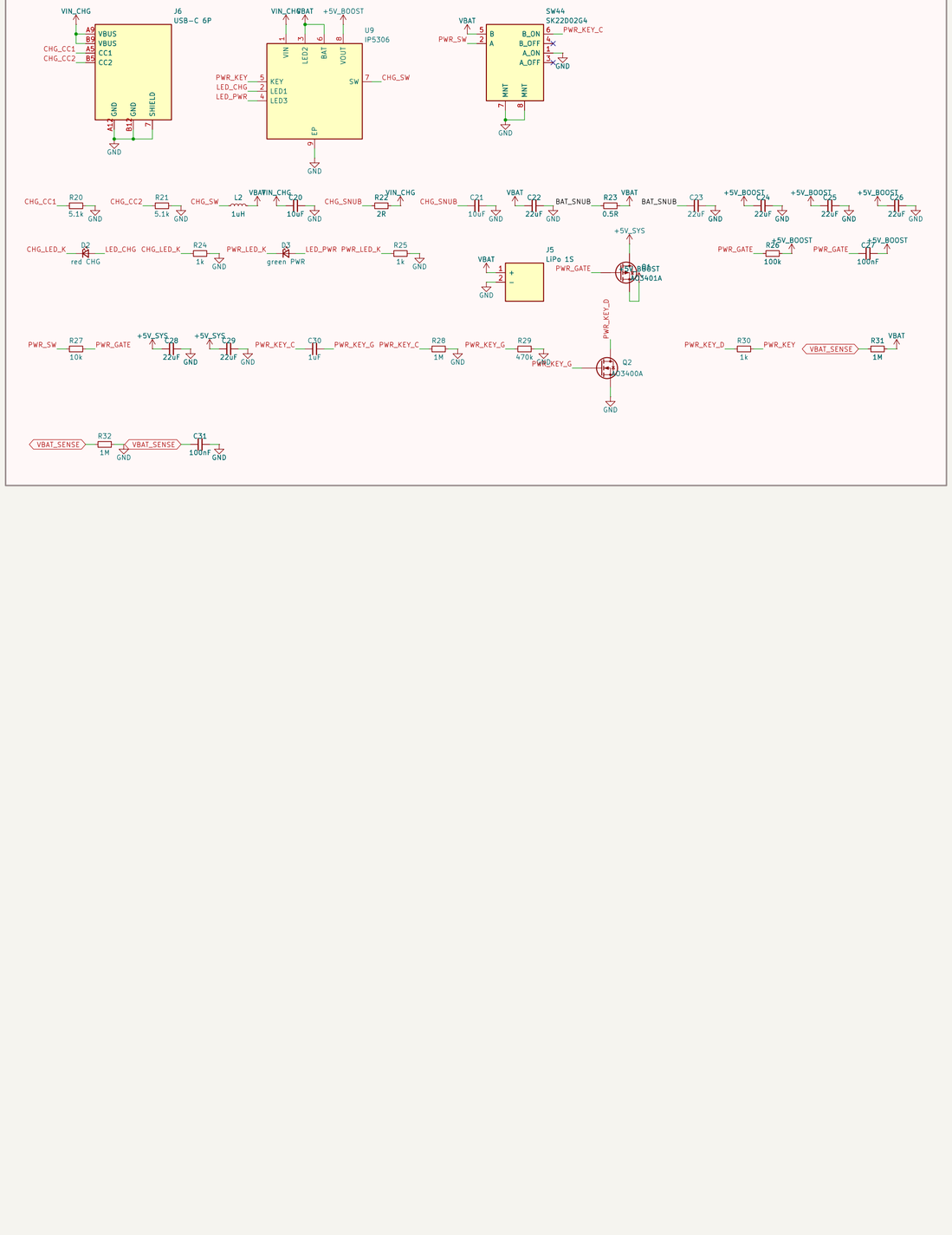
Title: Zine Machine main board v3: Keys and scan chain

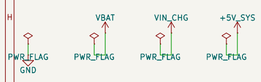
Size: A2	Date: 2026-11-06	**Rev: 3**
KiCad E.D.A. backplane-kicad-v10.99.0-bp.6		Id: 3/5

USB hub FE1.1s on USB1 -> 2 rear USB-A



Power: USB-C charge in, IP5306 1S charger/boost, slide switch + P-FET, LEDs





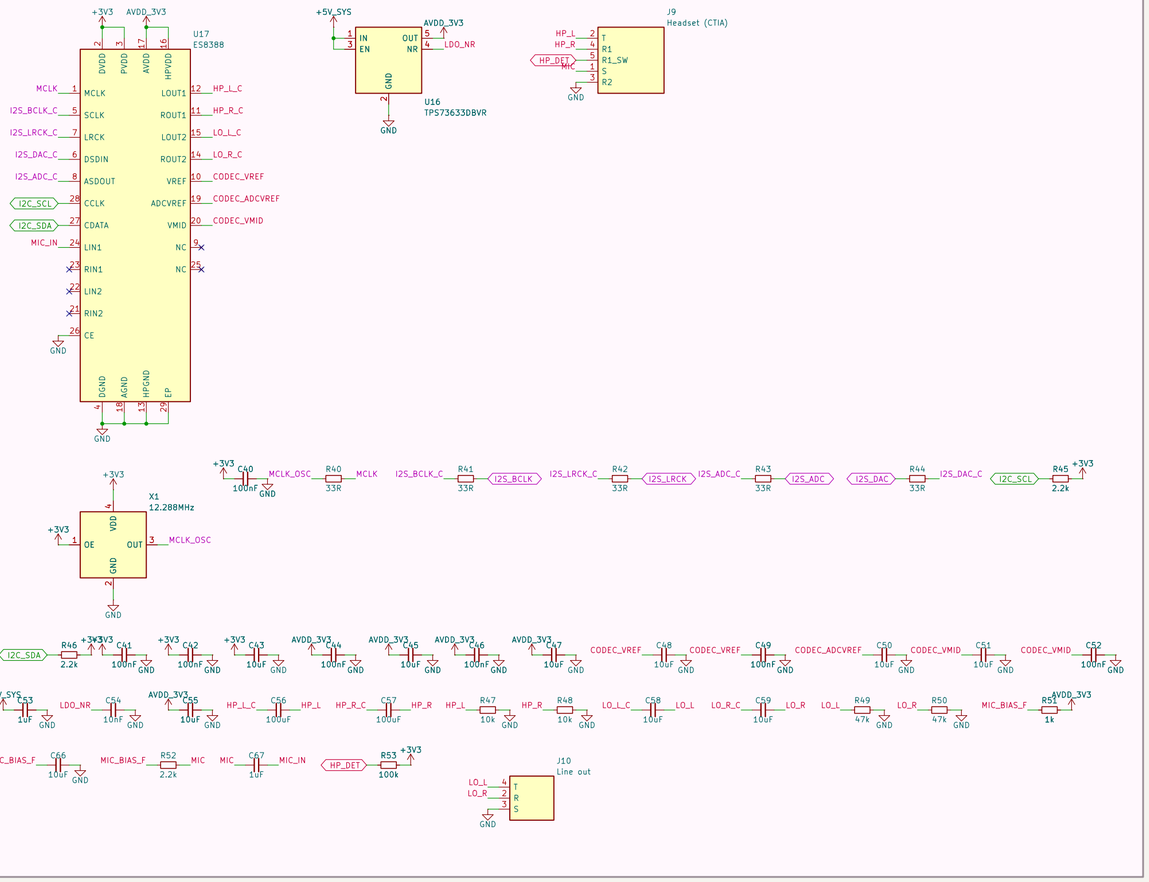
Generated by tools/zm_v3/gen_sch.py from design.py
bancouver40 + PocketBeagle 2 Industrial + 4.58in bar display

Sheet: /USB hub and power/
File: usb_power.kicad_sch

Title: Zine Machine main board v3: USB hub and power

Size: A2	Date: 2026-11-06	**Rev: 3**
KiCad E.D.A. backplane-kicad-v10.99.0-bp.6		Id: 4/5

Audio: ES8388 codec (I2S master), headset TRRS + line-out TRS



Generated by tools/zm_v3/gen_sch.py from design.py
bancouver40 + PocketBeagle 2 Industrial + 4.58in bar display

Sheet: /Audio/
File: audio.kicad_sch

Title: Zine Machine main board v3: Audio

Size: A2	Date: 2026-11-06	**Rev: 3**
KiCad E.D.A. backplane-kicad-v10.99.0-bp.6		Id: 5/5